MILITARY SPECIFICATION

MAINTENANCE PLATFORM SELF-PROPELLED, BOOM, LIFT, 40, 60 AND 80 FOOT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three sizes of one kind of commercial, self-propelled, multi-position, aerial work platform which will be called the unit.

1.2 Sizes. Units shall be either size 40, size 60 or size 80, as specified in the invitation for bids, request for proposals and contracts.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81	Tag, Shipping and Stock
PPP-B-601	Boxes, Wood, Cleated Plywood

MILITARY

MIL-C-104	Crates; Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-E-10062	Engine, Preparation for Shipment and Storage of
MIL-L-46010	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1730

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-162	Materials Handling Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Process
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Air Force

8140775	Requirements for the Design and Material to be used in the Design of Support Equipment (SE)

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC (ANSI)

ANSI A92.5 Boom - Supported Elevating Work Platforms

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018).

STEEL STRUCTURES PAINTING COUNCIL BULLETIN SSPC-SP6-60 No. 10

(Application for copies should be addressed to Steel Structures Painting Council, 4400 Fifth Ave, Pittsburgh, PA 15213).

2.1.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. This specification provides for first article inspection and approval (see 4.3 and 6.3).

3.2 Commercial Product. The unit shall be on the date of either the invitation for bids or request for proposals a standard commercial product of its manufacturer except for the engine spark arresting muffler (see 3.6.1), corrosion control finish (see 3.11) and operation instruction plate (see 3.13). The unit shall be of the manufacturer's latest production model.

3.3 Description. The unit shall be self-propelled, with forward and reverse speeds, gasoline engine driven, two axle, pneumatic tired, four wheel, two wheels driving, either two or four wheels steering. The unit shall be fitted with a powered turntable with telescopic boom terminating in a basket type work platform. The unit shall conform to the requirements of ANSI A92.5.

3.4 Dimensions.

3.4.1 Platform Heights. With boom fully elevated and extended, size 40, 60 and 80 units shall have platform height of not less than 40, 60 and 80 feet respectively measured from the ground plane to the work platform standing surface. With boom fully depressed and retracted, units shall have a platform height of not more than 44 inches measured from the ground plane to the work platform standing surface.

3.4.2 Other Platform Dimensions. The work platform shall have minimum dimensions of 3 feet by 5 feet.

3.5 Performance. The unit shall have a top speed forward on flat, level paved roadway of not less than 2.5 miles per hour. The unit shall be able to accomplish a flat, 15 percent grade, paved, roadway at not less than 1.5 miles per hour. The work platform boom turntable shall swing full circle continuously in both directions. With the turntable fixed, the work platform shall rotate not less than 80 degrees in both directions. The work platform shall be self-leveling with manual adjustment. The boom shall be infinitely extensible. The unit shall be designed and constructed for use down to minus 40 degrees Fahrenheit.

3.6 Details.

3.6.1 Engine. The manufacturer's standard commercial, gasoline-fueled, four stroke cycle engine shall be furnished. The engine shall be complete with a 12-volt starting, lighting and charging system. An air cleaner shall be provided. The power and speed rating of the engine shall be such that the operation of the unit under any of the requirements or conditions specified herein shall not exceed 90 percent of the engine manufacturer's published maximum horsepower. A fuel gage shall be provided. A spark arresting muffler shall be provided. If the muffler outlet is upward directed, then a stainless steel rain cap shall be installed. The manufacturer's standard commercial storage battery shall be furnished.

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3.6.2 Controls. All controls shall be dependable and shall not be erratic.

3.6.2.1 Platform. The work platform as a minimum shall be furnished with the following primary controls: Deadman, lift, steer, drive, forward/reverse, telescope, swing, platform leveling, drive/engine speed, ignition/start, hydraulic filter condition indicator, auxiliary power switch horn, light switches and control panel lights.

3.6.2.2 Ground. The ground control station shall as a minimum be furnished with the following secondary controls: Lift, swing telescope, auxiliary power, ignition/start, platform/ground/off control selector switch.

3.6.2.3 Other Controls. Other controls as needed shall be provided.

3.6.3 Work Platform. The work platform shall be provided with a 120 volt, 20 amp duplex outlet for powering of electric hand tools; the power sources for the outlet will be external to the unit. The work platform deck shall be clear - no support structure shall project through the platform deck.

3.6.4. Brakes. Service brakes shall be automatically applied when the unit is not in the drive mode. Brake release shall be hydraulic powered. Brakes shall hold on a 15 percent flat, smooth incline.

3.6.5 Standard Equipment. All standard equipment (for example: safety belts, hour meter) supplied to commercial customers shall be supplied with the unit.

3.6.6 Steering. Steering shall be hydraulic powered.

3.6.7 Holding Valves. Holding valves shall be provided in the hydraulic system to maintain work platform position in the event of loss of unit hydraulic pressure.

3.6.8 Lights. The unit shall be provided with the following lights: rotating amber beacon, head, red rear, back-up work platform adjustable flood, control panel (platform and ground) illumination.

3.7 General. The unit shall be suitable for its intended use. Structural elements shall be rigid and possess the requisite strength. Individual components shall be durable and reliable. Peculiar tools required to perform maintenance and repair shall be furnished with each unit. The unit shall be free of defects affecting strength, durability and reliability.

3.8 Completeness. The unit shall be furnished complete and serviced except for fuel; ready to perform its intended function when fueled.

3.9 Accessibility. Engine and hydraulic system components shall be readily accessible for servicing and repair.

3.10 Safety. Safety shall be a primary design, construction and operational consideration.

3.10.1 Bursting Safety Factors. All hydraulic components whose failure could result in free and unrestricted motion of the boom shall have a minimum bursting strength of at least four times the operating pressure for which the

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system is designed. All hydraulic components normally rated according to bursting strength (such as hose, tubing and fittings) shall have minimum bursting strength of at least three times the operating pressure for which the system is designed. All hydraulic components normally rated according to performance criteria, such as rated flow and pressure, life cycles, pressure drop, rpm torque and speed, shall have a minimum bursting strength of at least two times the operating pressure for which the system is designed. Such components generally include pumps, motors, directional controls, and similar functional components.

3.11 Corrosion Control.

3.11.1 Component and Procedures. The provisions of Air Force Drawing 8140775, Section III, paragraphs A (Exhaust Systems, C (Hinges), D (Latches), E (Rustproofing), G (Closed Hollow Members), H (Fuel Tanks), I (Spot/Tack Welded Assemblies, Bolted/Riveted Assemblies), J (Drainage) and L (Non-Standard Hardware) shall be satisfied in the respective components and procedures.

3.11.2 Finish. Abrasive blast ferrous materials to minimum of a near white finish per the Steel Structures Painting Council SSPC-SP6-60 No. 10 to remove all mill scale, ruts, oil, oxides, and soils. Remove all blasting residue. Condition surface with DuPont 5717s per manufacturer's instructions. Apply zinc phosphate treatment, DuPont 224S or equivalent, per manufacturer's instructions. Aluminum components will not be blasted. Prior to painting, they shall be thoroughly cleaned and given an appropriate deoxidizing and conversion coating treatment. Apply two coats of DuPont Color epoxy primer; primer number 825-S, activator number 825-S per manufacturer's instructions. Each coat shall be .8 to 1.0 mils dry film thickness for a total dry film thickness of 1.6 to 2.0 mils. Time between coats will be per primer manufacturer's recommendations for film thickness and environmental conditions (approx 1 to 4 hours). Minimum time before top coating will be from 2 to 6 hours, depending upon primer manufacturer's recommendations for film thickness and environmental conditions. Apply two coats of DuPont Imron, polyurethane topcoat, manufacturer's color 6561U, each coat being .8 to 1.0 mils dry film thickness. Total topcoat dry film thickness 1.6 to 2.0 mils. Time between coats shall be at least 15 minutes but not more than 2 hours. Paint manufacturer's recommendations for low or high temperature and humidity application techniques shall be followed if applicable. Areas to be cleaned and painted with the Imron system will include all areas normally painted with standard finishes to include the interior and underside of the cabinet enclosure and chassis. Particular attention shall be given to weld and seam areas. If previously painted, the engine, hydraulic components, etc within the cabinet enclosure do not require either blast cleaning or the Imron paint system. Surfaces not painted during normal productions due to sliding wear will be coated with dry film lubricant MIL-L-46010.

3.12 Identical Products. All units of the same size furnished under any one contract shall be identical and shall be identified by the same manufacturer's part number.

3.13 Operation Instruction Plate. The manufacturer shall attach a plate or plates in a readily accessible and clearly visible area stating: make, model and manufacturer's serial number, rated capacity, platform height, maximum hydraulic operating pressures, caution or restrictions in operations and operating instructions. The plates as a minimum shall prominently display "DO NOT USE ON OR NEAR HIGH VOLTAGE LINES."

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification. The inspection and testing of the unit shall be classified as follows:

a. First Article (see 4.3)

b. Acceptance (see 4.4)

4.3 First Article.

4.3.1 First Article inspection. One unit of each size being procured shall be subjected to the examination and tests specified in 4.5, Test Method. The sample shall be identical in all respects to the units to be delivered.

4.3.2 First Article Test Report. Upon completion of the first article examination and tests, a first article test report for each size unit procured shall be prepared in accordance with MIL-STD-831. The following shall be incorporated as a part of the report:

a. Manufacturer's dimensioned, either design or manufacturing, drawings confirming the dimensions of para 3.4.

b. Manufacturer's sales brochures, specification sheet, etc, covering the item he proposes to furnish.

c. Evidence satisfactory to the procuring activity that the unit complies with ANSI A92.5.

d. Evidence satisfactory to the procuring activity that the unit complies with the horsepower requirements of para 3.6.1 herein.

e. Evidence satisfactory to the procuring activity that the unit complies with the bursting safety factor requirement herein.

f. Evidence satisfactory to the procuring activity that the unit complies with the Corrosion Control requirement specified herein.

g. Manufacturer's standard commercial acceptance inspection and test procedures.

4.3.3 Test Report Approval. The first article test report shall be submitted to the procuring activity for review and approval/disapproval. No unit shall be delivered to the Government prior to notification to the contractor of procuring activity approval of the test report.

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4.4 Acceptance. Acceptance inspection of each production item shall consist of the manufacturer's standard commercial inspection and test procedures.

4.5 Test Method

4.5.1 Examination. The unit shall be examined to determine compliance with the requirements herein. The examination shall be performed using a contractor developed check list which lists all requirements of Section 3 herein not tested for, method of confirmation and results obtained. Engineering (not sales) drawings, either manufacturing or design, shall be reviewed to show compliance with the dimensions specified in para 3.4 herein. Manufacturer's sales material shall be reviewed to demonstrate that item to be supplied is a current standard commercial product. Evidence shall be examined confirming that the unit complies with the requirements of ANSI A92.5. Evidence shall be examined that the unit complies with the horsepower requirement of para 3.6.1 herein. Evidence shall be examined confirming that the unit complies with the bursting strength factor specified herein. Evidence shall be examined confirming that the unit complies with the Corrosion Control requirement specified herein.

4.5.2 Road Tests. The unit shall be tested to demonstrate a level road speed of not less than 2.5 miles per hour. The unit shall be tested to show a speed up a 15 percent incline of not less than 1.5 mph. Reverse speed shall be demonstrated. Driving and steering from both ground and platform control stations shall be demonstrated.

4.5.3 Brake Test. The unit shall be tested to show that the brakes hold on a flat, smooth 15 degree incline. Automatic application of brakes when unit is not in drive mode shall be demonstrated. Hydraulic release of brakes shall be demonstrated.

4.5.4 Controls Test. All controls shall be demonstrated.

4.5.5 Holding Valves Test. Effectiveness of the holding valve with the work platform at its maximum elevation and reach and with the maximum rated work platform load shall be demonstrated.

4.5.6 Access Test. Ready access to the engine and hydraulic system components shall be demonstrated.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Disassembly and Matchmarking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage, loss or breakage in transit and to accomplish reduction in cube. Bolts, nuts, screws, pins, washers and other attaching parts removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary to facilitate reassembly, parts removed and mating parts shall be matchmarked. Matchmarking shall be printed on shipping tags conforming to UU-T-81, Type A. The printed tags shall be waterproofed in accordance with MIL-STD-129.

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5.1.2 Level A. The maintenance platform shall be preserved in accordance with MIL-P-116, Method I. All bare or unpainted metal surfaces subject to rust or corrosion shall be coated with type P-1 preservative in accordance with MIL-P-116. The hydraulic systems shall be preserved in accordance with MIL-STD-162. Engine shall be preserved in accordance with MIL-E-10062.

5.1.3 Commercial. Maintenance platforms shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A & B. The maintenance platform preserved and packaged as specified in 5.1.2 shall be packed in a container conforming to MIL-C-104, Style A. All loose parts shall be packed in a container conforming to PPP-B-601, Type II which shall be secured within the MIL-C-104 Crate. Containers shall be closed and strapped in accordance with the appendix of the applicable specification.

5.2.2 Commercial. Maintenance platform preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129, or ASTM D 3951 as applicable.

5.4 Physical protection. All cushioning, blocking, bracing and bolting shall be in accordance with MIL-STD-1186.

6. NOTES

6.1 Intended Use. The unit is civil engineering equipment for use in the inspection and maintenance of electrical installations, heating and ventilating equipment overhead cranes and structures of large Air Force buildings such as aircraft hangars.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of each size, either 40, 60 or 80.

c. Levels of preservation, packaging and packing (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a standard production item from the contractor's current inventory. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

Custodian:
Army - AV
Navy - AS
Air Force - 99

Preparing Activity:
Air Force - 82

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-87973	Maintenance, Platform, Self-Propelled Boom, 40, 60 and 80 Foot

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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